Now imagine a time, a few decades hence, when all such near-Earth asteroids are inventoried and their orbits compiled. Then, as Alan Harris of JPL, Greg Canavan of the Los Alamos National Laboratory, Ostro, and I have shown, it might take only a year to select a suitable object, alter its orbit, and send it crashing into the Earth with cataclysmic effect.

The technology required—large optical telescopes, sensitive detectors, rocket propulsion systems able to lift a few tons of payload and make precise rendezvous in nearby space, and thermonuclear weapons—all exist today. Improvements in all but perhaps the last can be confidently expected. If we're not careful, many nations may have these capabilities in the next few decades. What kind of world will we then have made?

We have a tendency to minimize the dangers of new technologies. A year before the Chernobyl disaster, a Soviet nuclear power industry deputy minister was asked about the safety of Soviet reactors, and chose Chernobyl as a particularly safe site. The average waiting time to disaster, he confidently estimated, was a hundred thousand years. Less than a year later … devastation. Similar reassurances were provided by NASA contractors the year before the *Challenger* disaster: You would have to wait ten thousand years, they estimated, for a catastrophic failure of the shuttle. One year later … heartbreak.

Chlorofluorocarbons (CFCs) were developed specifically as a completely safe refrigerant—to replace ammonia and other refrigerants that, on leaking out, had caused illness and some deaths. Chemically inert, nontoxic (in ordinary concentrations), odorless, tasteless, nonallergenic, nonflammable, CFCs represent a brilliant technical solution to a well-defined practical problem. They found uses in many other industries besides refrigeration and air conditioning. But, as I described above, the chemists who developed CFCs overlooked one essential fact—that the molecules' very inertness guarantees that they are circulated to stratospheric altitudes and there cracked open by sunlight, releasing chlorine atoms which then attack the protective ozone layer. Due to the work of a few scientists, the dangers may have been recognized and